

Exercise 7: Financial Forecasting

1. Aim

To develop a **Financial Forecasting Tool** using a recursive algorithm in Java to predict future values based on a constant annual growth rate.

2. Problem Understanding

Financial forecasting is the process of estimating future financial values based on historical data and expected growth rates.

Recursion is a programming technique where a method calls itself to solve a problem by breaking it into smaller subproblems. It is useful for problems that have a repetitive structure.

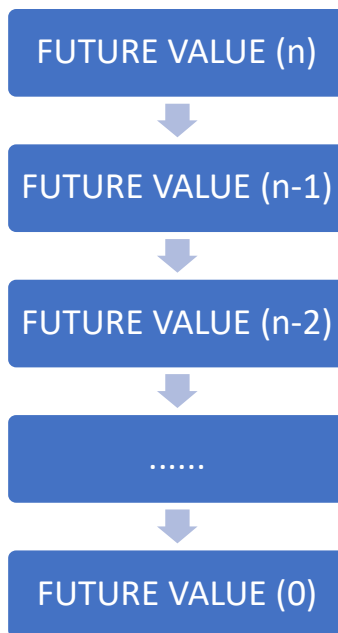
For this exercise, the future value is calculated by applying the growth rate recursively for a given number of years.

3. Concept of Recursion

A recursive function consists of:

- **Base Case:** Stops the recursion.
- **Recursive Case:** Calls the function again with a smaller problem.

Example:



The recursion stops when the number of years becomes zero.

4. Algorithm

1. Read the initial investment amount.
2. Read the annual growth rate.
3. Read the number of years.
4. If the number of years is zero, return the current value.
5. Otherwise, increase the value by the growth rate.
6. Recursively calculate the value for the remaining years.
7. Display the predicted future value.

5. Program/Output:

```
J Exercise7.java > ...
1 public class Exercise7 {
2     public static double foreCast(double currentValue, double growthRate, int years){
3         if(years==0)
4             return currentValue;
5         return foreCast(currentValue*(growthRate+1), growthRate, years-1);
6     }
7     Run | Debug
8     public static void main(String[] args) {
9         double currentValue = 100000;
10        double growthRate = 0.1;
11        int year =5;
12        double futureValue = foreCast(currentValue, growthRate, year);
13        System.out.println("Current Value: Rs."+currentValue);
14        System.out.println("GrowthRate: "+growthRate*100+"%");
15        System.out.println("Years: "+year);
16        System.out.printf(format: "Predicted Future Value: Rs. %.2f%n ",futureValue);
17    }
18 }
```

```
PS C:\Users\Krishnavishwa\OneDrive\Documents\Material\cognizant\Algorithm> & 'C:\
navishwa\AppData\Roaming\Code\User\workspaceStorage\84f21bc9d5619c8f027c7d00daccf3
Current Value: Rs.100000.0
GrowthRate: 10.0%
Years: 5
Predicted Future Value: Rs. 161051.00

PS C:\Users\Krishnavishwa\OneDrive\Documents\Material\cognizant\Algorithm>
```

6. Time Complexity Analysis

Operation	Time Complexity
Recursive Forecast	O(n)

Time Complexity: $O(n)$, **Space Complexity:** $O(n)$ (due to the recursion call stack)

7. Result

Thus, the **Financial Forecasting Tool** was successfully implemented using recursion in Java. The program predicts future financial values by repeatedly applying the annual growth rate. The recursive approach has a time complexity of **$O(n)$** and demonstrates how recursion can simplify repetitive calculations, although iterative or mathematical approaches are more efficient for large inputs.